DS HW SET 1-2025115112

1) Sum of 2 matrices

Code

```
#include <stdio.h>

int main()

{

    int m, n;

    printf("Enter size of 1st matrix: ");

    scanf("%d %d", &m, &n);


    int a[m][n], b[m][n], sum[m][n];

    int i, j;


    printf("Enter elements of first matrix: \n");

    for (i = 0; i < m; i++)

    {

        for (j = 0; j < n; j++)

        {

            scanf("%d", &a[i][j]);

        }

    }


    printf("Enter elements of second matrix: \n");

    for (i = 0; i < m; i++)

    {

        for (j = 0; j < n; j++)

        {

            scanf("%d", &b[i][j]);

        }

    }


    printf("\nSum of 2 matrices:\n");

    for (int i = 0; i < m; i++)

    {
```

```
        for (int j = 0; j < n; j++)
        {
            sum[i][j] = a[i][j] + b[i][j];
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:

Enter size of matrix: 2 2

Enter elements of 1st matrix:

1 2

3 4

Enter elements of 2nd matrix:

5 6

7 8

Sum of two matrices

6 8

10 12

2) Read n values in an array and display in reverse order

Code

```
#include <stdio.h>

int main()

{

   int n = 5;

   int arr[5] = {10, 20, 30, 40, 50};


   printf("Elements in reverse order: \n");

   for (int i = n - 1; i >= 0; i--)

   {

      printf("%d ", arr[i]);

   }

   printf("\n");


   return 0;

}
```

Output:

Elements in reverse order:

50 40 30 20 10

3) Count the total number of duplicate elements in an array

Code

```
#include <stdio.h>


int main()
{
    int arr[] = {1, 2, 2, 3, 4, 4, 4, 5};
    int n = sizeof(arr) / sizeof(arr[0]);
    int dup = 0;

    for (int i = 0; i < n; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            if (arr[i] == arr[j])
            {
                dup++;
                break;
            }
        }
    }

    printf("Total no. of duplicate elements: %d\n", dup);

    return 0;
}
```

Output:

Total no. of duplicate elements: 2

4) Print unique elements in an array

Code

```
#include <stdio.h>


int main()
{
    int arr[] = {1, 2, 2, 3, 4, 4, 5};
    int n = sizeof(arr) / sizeof(arr[0]);

    printf("Unique Elements: ");
    for (int i = 0; i < n; i++)
    {
        int c = 0;
        for (int j = 0; j < n; j++)
        {
            if (arr[i] == arr[j])
            {
                c++;
            }
        }
        if (c == 1)
        {
            printf("%d ", arr[i]);
        }
    }
    printf("\n");

    return 0;
}
```

Output:

Unique Elements: 1 3 5

5) Insert an element into a sorted array

Code

```
#include <stdio.h>

int main()
{
    int arr[10] = {10, 20, 30, 50, 60};
    int n = 5, key = 40, i;

    for (i = n - 1; (i >= 0 && arr[i] > key); i--)
    {
        arr[i + 1] = arr[i];
    }
    arr[i + 1] = key;
    n++;

    printf("Array after insertion: ");
    for (i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }
    printf("\n");

    return 0;
}
```

Output:

Array after insertion: 10 20 30 40 50 60

6) Insert an element into an unsorted array

Code

```
#include <stdio.h>


int main()
{
    int arr[10] = {10, 25, 5, 12};
    int n = 4, key = 99, pos = 2;


    for (int i = n; i >= pos; i--)
    {
        arr[i] = arr[i - 1];
    }
    arr[pos] = key;
    n++;


    printf("Array after insertion: ");
    for (int i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }
    printf("\n");


    return 0;
}
```

Output:

Array after insertion: 10 25 99 5 12

7) Matrix Multiplication

Code:

```
#include <stdio.h>


int main() {
   int m1,n1,m2,n2;
   printf("Enter size of matrix A ");
   scanf("%d %d", &m1,&n1);

   printf("Enter size of matrix B ");
   scanf("%d %d", &m2,&n2);

   if(n1!=m2)
   {
      printf("Invalid");
      return 1;
   }
   else
   {
      int i,j,k;
      int A[m1][n1],B[m2][n2];

      printf("Enter elements for A:\n ");
      for(i=0;i<m1;i++)
      {
         for(j=0;j<n1;j++)
         {
            scanf("%d",&A[i][j]);
         }
      }



   printf("Enter elements for B:\n ");
      for(i=0;i<m2;i++)
```

```
    {
        for(j=0;j<n2;j++)
        {
            scanf("%d",&B[i][j]);
        }
    }


int M[m1][n2];
for(i=0;i<m1;i++)
    {
        for(j=0;j<n2;j++)
        {
            M[i][j]=0;
            for(k=0;k<n1;k++)
            {
                M[i][j]+=A[i][k]*B[k][j];
            }
        }
    }


for(i=0;i<m1;i++)
    {
        for(j=0;j<n2;j++)
        {
            printf("%d ",M[i][j]);
        }
        printf("\n");
    }
}
```

```
    return 0;
}
```

Output:

Enter size of matrix A: 3 2

Enter size of matrix B: 2 3

Enter elements for A:

1 2

2 1

3 0

Enter elements for B:

1 2 3

1 0 0

Result:

3 2 3

3 4 6

3 6 9

8) Matrix transpose

Code:

```
#include <stdio.h>


int main() {
   int m,n;
   printf("Enter size of matrix: ");
   scanf("%d %d", &m,&n);


      int i,j;
      int A[m][n],T[n][m];


      printf("Enter elements for A\n");
      for(i=0;i<m;i++)
      {
         for(j=0;j<n;j++)
         {
            scanf("%d",&A[i][j]);
         }
      }




   for(i=0;i<n;i++)
      {
         for(j=0;j<m;j++)
         {
            T[i][j]=A[j][i];
         }
      }

   Printf("Transpose of A\n");
   for(i=0;i<n;i++)
      {
```

```
            for(j=0;j<m;j++)
            {
                printf("%d ",T[i][j]);
            }
            printf("\n");
        }
    return 0;
}
```

Output:

Enter size of matrix: 2 3

Enter elements for A

1 2 3

4 5 6

Transpose of A:

1 4

2 5

3 6

9) Right diagonal sum

Code:

```
#include <stdio.h>
int main()
{
    int m;
    printf("Enter size of square matrix (m x m): ");
    scanf("%d", &m);

    int i, j;
    int A[m][m];

    printf("Enter elements for matrix A:\n");
    for (i = 0; i < m; i++) {
        for (j = 0; j < m; j++) {
            scanf("%d", &A[i][j]);
        }
    }

    int s = 0;
    for (i = 0; i < m; i++) {
        s += A[i][m - 1 - i];
    }

    printf("Sum of Right Diagonal Elements: %d\n", s);

    return 0;
}
```

Output:

```
Enter size of square matrix (m x m): 2
Enter elements for matrix A:
1 2
4 7
Sum of Right Diagonal Elements: 6
```

10) Check for identity matrix

Code:

```
#include <stdio.h>


int main() {
    int m;
    printf("Enter size of square matrix (m x m): ");
    scanf("%d", &m);


    int i, j;
    int A[m][m];


    printf("Enter elements for matrix A:\n");
    for (i = 0; i < m; i++) {
        for (j = 0; j < m; j++) {
            scanf("%d", &A[i][j]);
        }
    }


    int f = 1; // Flag: 1 = true, 0 = false


    for (i = 0; i < m; i++)
{
        for (j = 0; j < m; j++)
            {
                    if ((i == j && A[i][j] != 1) || (i != j && A[i][j] != 0))
                {
                    f = 0;
                    break;
                    }
                }
                if (f == 0)
            {
                    break;
```

```
        }
}

  if (f== 1) {
    printf("The matrix is an Identity Matrix.\n");
  } else {
    printf("The matrix is NOT an Identity Matrix.\n");
  }

  return 0;
}
```

Output:

Enter size of square matrix (m x m): 3

Enter elements for matrix A:

1 0 0

0 1 0

0 0 1

The matrix is an Identity Matrix.

Enter size of square matrix (m x m): 3

Enter elements for matrix A:

1 0 0

0 5 0

0 0 1

The matrix is NOT an Identity Matrix.

11)To find no. of vowels, consonants, digits and white spaces

Code:

```
// Online C compiler to run C program online
#include <stdio.h>

int main()
{
    char str[50];
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    int v = 0, c = 0, b = 0, i = 0;

    while (str[i] != '\0')
    {
        char x = str[i];

        if (x == 'a' || x == 'e' || x == 'i' || x == 'o' || x == 'u' ||x == 'A' || x == 'E' || x == 'I' || x == 'O' || x == 'U')
        {
            v++;
        }
        else if ((x >= 'a' && x <= 'z') || (x >= 'A' && x <= 'Z'))
        {
            c++;
        }
        else if (x == ' ')
        {
            b++;
        }
        i++;
    }

    printf("Number of Consonants: %d\nNumber of Vowels: %d\nNumber of blank spaces: %d\n", c, v, b);
```

```
    return 0;
}
```

Output:

Enter a string: Hello World

Number of Consonants: 7

Number of Vowels: 3

Number of blank spaces: 1

12) Frequency of a character in a string

Code:

```
#include <stdio.h>


int main()

{

    char str[100];

    char ch;

    int count = 0, i = 0;


    printf("Enter a string: ");

    fgets(str,sizeof(str),stdin);


    printf("Enter a character to find its frequency: ");

    scanf("%c", &ch);


    while (str[i] != '\0')

    {

        if (str[i] == ch)

        {

            count++;

        }

        i++;

    }


    printf("Frequency of '%c' = %d\n", ch, count);


    return 0;

}
```

Output:

Enter a string: hello world

Enter a character to find its frequency: l

Frequency of 'l' = 3

13) Count words in a string

Code:

```c
#include <stdio.h>

int main()
{
    char str[100];
    char ch;
    int w = 0, i = 0;

    printf("Enter a string: ");
    fgets(str,sizeof(str),stdin);

    while (str[i] != '\0')
    {
        if (str[i] == ' '||str[i+1]=='\0')
        {
            w++;
        }
        i++;
    }

    printf("No. of words: %d\n", w);

    return 0;
}
```

Output:

Enter a string: Hello C programming is fun

No. of words: 5

14) Sort string array

Code:

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, j;
    char temp;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    int len = 0;
    while (str[len] != '\0' && str[len] != '\n')
    {
        len++;
    }

    for (i = 0; i < len - 1; i++)
    {
        for (j = 0; j < len - 1 - i; j++)
        {
            if (str[j] > str[j + 1])
            {
                temp = str[j];
                str[j] = str[j + 1];
                str[j + 1] = temp;
            }
        }
    }

    str[len] = '\0';
```

```
    printf("Sorted string: %s\n", str);


    return 0;
}
```

Output:

Enter a string: hello

Sorted string: ehllo

15) Toggle case of a sentence (lowercase to uppercase & vice versa)

Code:

```
#include <stdio.h>


int main()
{
    char str[100];
    int i = 0;

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0')
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
        else if (str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
        i++;
    }

    printf("Toggled sentence: %s", str);

    return 0;
}
```

Output:

Enter a sentence: HeLlO wOrLd!!

Toggled sentence: hElLo WoRlD!!